Choose a reconstruction by its conditional squared-error risk

Choose a reconstruction after observing y

$$m = E[z|y]$$

Unknown $z \in \mathbb{R}^d$; choose $a \in \mathbb{R}^d$ to reconstruct it.

Squared-error criterion

$$L(a, z) = \|z - a\|^2, \quad R(a|y) = E[L(a, z)|y]$$

Center the error at the conditional mean

$$r = z - m, \quad z - a = r + (m - a)$$

$$\text{align(center)}[\|z - a\|^2 = \|r\|^2 + 2r^T(m - a) + \|m - a\|^2]$$

- Choose a to minimize conditional expected squared error $R(a|y)$.
- Given y, a and m are fixed; the residual r remains random.